# Summary Notes: Cryptography

• Cryptography - Summary Notes (Secondary Level)**

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## 1. What is Cryptography?

• The science of keeping information **private**, **accurate**, and **authentic**.

• It does this by turning readable data (**plaintext**) into scrambled data (**ciphertext**) and back again using **algorithms** (called *ciphers*) and **keys**.

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## 2. Core Vocabulary

| Term | Simple definition |

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| **Plaintext** | The original, readable message. |

| **Ciphertext** | The scrambled version that looks meaningless without the key. |

| **Encryption** | The process that changes plaintext into ciphertext. |

| **Decryption** | The reverse process that restores ciphertext to plaintext. |

| **Key** | A secret number or string that tells the cipher how to scramble or unscramble. |

| **Hash function** | A one-way tool that turns any data into a short, fixed-size "digest". It cannot be reversed and is used to check that data hasn't been altered. |

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## 3. Symmetric vs. Asymmetric Cryptography

| Feature | Symmetric (same key) | Asymmetric (key pair) |

|---|---|---|

| **Key usage** | One secret key for both encryption & decryption. | Public key (anyone can use) + private key (kept secret). |

| **Speed** | Very fast - good for large amounts of data. | Slower - usually only for small data (e.g., exchanging a symmetric key). |

| **Key distribution problem** | Must share the secret key securely first. | No need to share the private key; public key can be posted openly. |

| **Common algorithms** | **AES** (Advanced Encryption Standard) - 128/192/256-bit keys. | **RSA** - security based on factoring large numbers. <br>**ECC** - elliptic-curve cryptography, same security as RSA with

shorter keys. |

| **Typical use** | Encrypting files, video streams, VPN traffic. | Setting up a secure channel (TLS/HTTPS), digital signatures. |

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## 4. Historical Milestones & Important Algorithms

| Era / Event | Why it matters |

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| **Caesar Cipher (≈ 50 BC)** | First recorded systematic substitution - each letter shifted by a fixed amount. |

| **Enigma Machine (WWII)** | Mechanical rotor system; its eventual break showed the power of systematic cryptanalysis. |

| **DES (1977)** | First widely adopted government standard; 56-bit key now too short for modern computers. |

| **AES (2001)** | Replaced DES; uses 128-, 192-, or 256-bit keys; currently the global standard for symmetric encryption. |

| **RSA (1977)** | First practical public-key system; enables secure key exchange and digital signatures. |

| **ECC (2000s)** | Provides RSA-level security with much smaller keys - ideal for phones and IoT devices. |

| **SHA-256 (2001)** | A secure hash algorithm that creates a 256-bit digest; used for file integrity and blockchain. |

| **Post-Quantum Cryptography (today)** | New algorithms (e.g., **CRYSTALS-Kyber**) being prepared for a future where quantum computers could break RSA/ECC. |

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## 5. Everyday Applications

• **HTTPS / TLS** - Web browsers use an asymmetric handshake to agree on a fast symmetric session key, then encrypt all traffic.

• **End-to-end encrypted messaging** (Signal, WhatsApp) - Combine asymmetric key exchange, symmetric encryption, and *forward secrecy* so only the two participants can read messages.

• **Digital signatures** - A private key creates a signature; anyone can verify it with the corresponding public key, proving authorship and integrity.

• **Certificate Authorities (CAs)** - Trusted organisations that issue digital certificates linking a public key to a real-world identity, forming the backbone of the Public Key Infrastructure (PKI).

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## 6. Threats & Future Directions

• **Brute-force attacks** - Trying every possible key; become realistic when keys are too short (e.g., < 128 bits).

• **Cryptanalysis** - Uses mathematics, statistical patterns, side-channel data (like power use), or future quantum computers to uncover keys.

• **Quantum computers** - Could solve the hard problems RSA and ECC rely on; motivates research into post-quantum algorithms.

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## 7. Quick Review - Key Takeaways

1. **Cryptography** turns plaintext ↔ ciphertext using algorithms and keys.

2. **Symmetric** = one secret key, **fast**, but needs a safe way to share the key.

3. **Asymmetric** = public-/private-key pair, solves key-distribution, used for handshakes and signatures.

4. Modern standards: **AES** (symmetric), **RSA/ECC** (asymmetric), **SHA-256** (hash).

5. Real-world systems blend both types to protect web traffic, messages, and software updates.

6. Ongoing threats (brute force, quantum computers) drive the move to longer keys and post-quantum algorithms.

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## 8. Multiple-Choice Questions (10)

| # | Question | Options (A-D) | Correct |

|---|----------|---------------|---------|

| 1 | Which term describes the original, readable message before encryption? | A) Ciphertext B) Plaintext C) Hash D) Key | B |

| 2 | In symmetric-key cryptography, the same key is used for ... | A) Encryption only B) Decryption only C) Both encryption and decryption D) Neither | C |

| 3 | Which algorithm is a **hash** function, not an encryption algorithm? | A) AES B) RSA C) SHA-256 D) ECC | C |

| 4 | The main advantage of asymmetric cryptography over symmetric cryptography is: | A) Faster encryption of large files B) No need to share a secret key C) Uses shorter keys D) It never needs a public key | B |

| 5 | Which of the following is **not** a symmetric-key algorithm? | A) AES B) DES C) RSA D) Blowfish | C |

| 6 | What does "forward secrecy" protect against? | A) Someone stealing a private key B) An attacker decrypting past messages after obtaining a key C) Loss of internet connection D) Hash collisions | B |

| 7 | The security of RSA relies on the difficulty of: | A) Solving discrete logarithms B) Factoring large prime numbers C) Computing elliptic-curve points D) Finding hash collisions | B |

| 8 | Which historical device used rotating wheels to create a polyalphabetic substitution? | A) Caesar Cipher B) Enigma Machine C) DES D) AES | B |

| 9 | A Certificate Authority (CA) primarily provides: | A) Symmetric keys for VPNs B) Digital certificates that bind public keys to identities C) Hash functions for file checksums D) Quantum-resistant algorithms | B |

|10| Which of the following is a post-quantum candidate algorithm? | A) RSA B) ECC C) CRYSTALS-Kyber D) SHA-1 | C |

• Answer Key:** 1-B, 2-C, 3-C, 4-B, 5-C, 6-B, 7-B, 8-B, 9-B, 10-C.